

Detection of Flight Anomalies during Landing

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Go around maneuvers, which are aborted landing approaches, requiring pilots to abandon approach to the runway and climb, are the most complex and critical parts of commercial flight. Timely detection of anomalies during landing is imperative to aviation safety. By using the NASA DASHlink Curated 4-Class Anomaly Detection Dataset, which contains about 99,000+ flights each containing data across 160 seconds and 20 sensor features, we trained multiple machine learning models to detect if these anomalies are present. For the baseline model, we used a Random Forest classifier. We then moved to a 1D Convolutional Neural Network (CNN) with increasing complexity. The final model was a 1D CNN that achieved a test accuracy of 97.62%, recall of 87.5%, and precision of 88.9%. These results demonstrate that these models can reliably detect subtle flight anomalies, such as high airspeed, a high approach path, or late flap deployment, using time series data from the plane, ultimately suggesting strong potential for deployment in real-time Flight Data Monitoring systems.

I. Introduction

Aviation safety is one of the highest priority in modern transportation. Despite significant advances in aircraft design, safety, and air traffic management, anomalies during critical flight phases like landing are a key contributing factor in aviation incidents and accidents.

This paper investigates 1D CNNs for flight anomaly detection during landing. We used the NASA DASHlink Curated 4-Class Anomaly Detection Dataset, which contains 99,000+ flights and labels them as nominal flights or anomaly flights, which are either speed high, path high, or flaps late. A Random Forest classifier established a baseline, followed by a progression of CNN models.

The primary objective is to determine whether CNNs can achieve a high anomaly recall, which is correctly identifying actual anomalies. Results show that the CNN achieved 87.5% recall and 0.9868 ROC-AUC. A website was also developed to demonstrate and show real deployment of the model.

II. Background

A go-around is performed when the flight crew determines that conditions are unsafe to continue an approach and landing. Common triggers include unstable approaches, runway incursions, or weather. In our model, we focus on

unstable approaches, looking for excessive approach airspeed, an above normal glidepath, or delayed flap deployment. By looking at the current flight data, our model predicts if the flight is either nominal or anomalous, which then suggests if they should perform a go around.

Convolutional Neural Networks, originally developed for image recognition, also work well for time series classification. A 1D CNN applies learned filters along the time axis to extract local temporal patterns across input channels. Residual connections allow gradients to flow through deep networks without degradation, enabling stable training of deep architectures. Dilated causal convolutions extend the temporal receptive field exponentially with dilation rate, allowing efficient capture of long range dependencies without increasing filter size.

III. Dataset

The study uses the NASA DASHlink Curated 4-Class Anomaly Detection Dataset, derived from real commercial aviation flight data recorder recordings. The dataset contains about 99,000+ flights, each containing 160 seconds of data. Each flight is labeled as 0 (nominal), 1 (speed high), 2 (path high), or 3 (flaps late). For this study, we combined 1, 2, and 3 as an anomaly, creating a binary classification. The dataset contains about 90% nominal flights and 10% anomalies. All 20 features available in the dataset are used as inputs to the CNN models. These range from flight measurements such as airspeed and altitude, control surface positions such as the rudder and flap, navigation parameters such as heading and glideslope deviation, and engine parameters such as core speed. Table 1 provides a full listing of all 20 features.

A. Data Preprocessing

For the CNN models, the dataset is split randomly into training (72.25%), validation (12.75%), and test (15%). The Random Forest uses an 80/20 train-test split.

IV. Methodology

For the baseline model, we used a Random Forest classifier with 100 decision trees. Each flight segment was flattened into a single vector using four selected features — airspeed, localizer deviation, core speed, and pitch angle — across all 160 timestamps. We then moved to a 1D CNN, which could learn temporal patterns directly from the raw data rather than relying on flattened feature vectors. The final model was an advanced residual CNN designed specifically for time series. It starts with a Gaussian noise layer, then an initial Conv1D layer, followed by a residual block.

V. Results and Discussion

Table 2 summarizes the test set results for all four models.

The Random Forest had the weakest anomaly detection performance, missing 50% of actual anomalies despite

Table 1 DASHlink dataset features

Index	Feature Name	Unit	Description
0	Aileron LH	degrees	Left aileron position
1	Aileron RH	degrees	Right aileron position
2	Angle of Attack	degrees	Corrected angle of attack
3	Altitude	feet	Barometric corrected altitude
4	Airspeed	knots	Computed airspeed
5	Selected Course	degrees	Autopilot selected course
6	Drift Angle	degrees	Heading vs. track angle
7	Elevator Left	degrees	Left elevator position
8	Flap Position	discrete	Trailing edge flap deployment
9	Glideslope Dev.	%	Deviation from glideslope
10	Selected Heading	degrees	Autopilot desired heading
11	Localizer Dev.	%	Runway axis lateral deviation
12	Core Speed Avg.	%	Average engine compressor speed
13	Total Pressure	millibar	Total air pressure
14	Pitch Angle	degrees	Pitch angle
15	Roll Angle	degrees	Roll angle
16	Rudder Position	degrees	Rudder position
17	True Heading	degrees	True heading
18	Vertical Accel.	G	Vertical acceleration
19	Wind Speed	knots	Wind speed

Table 2 Test set performance comparison across all models

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	93.85%	82.9%	50.0%	62.4%	—	—
1-Layer CNN	96.90%	84.0%	85.9%	84.9%	0.9823	0.9098
2-Layer CNN	97.31%	87.1%	86.4%	86.7%	0.9841	0.9193
Residual CNN	97.62%	88.9%	87.5%	88.2%	0.9868	0.9392

having 93.85% overall accuracy. This shows why accuracy alone is misleading when classes are imbalanced. A model that always predicts nominal would still nearly get 90% accuracy.

All of the CNNs significantly improved on recall. The 1-layer CNN reached 85.9% recall and a ROC-AUC of 0.9823, as shown in Fig. 2 and Fig. 3. Adding a second convolutional layer bumped the recall to 86.4% and ROC-AUC to 0.9841, as shown in Fig. 4. The residual CNN hit 87.5% recall, 88.9% precision, and a ROC-AUC of 0.9858, as shown in Fig. 5.



heatmap_samples.png

Fig. 1 Random training samples visualized as heatmaps ($20 \text{ features} \times 160 \text{ timesteps}$). The bottom right sample in each grid is labeled anomaly; all others are nominal.

VI. Conclusion

Acknowledgments

References

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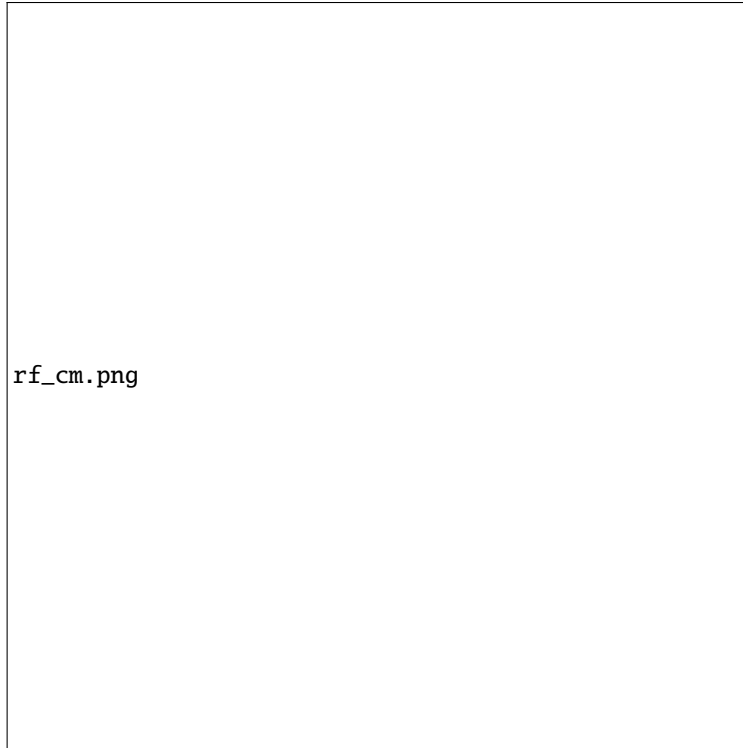


Fig. 2 Confusion matrix for the Random Forest baseline on the test set.

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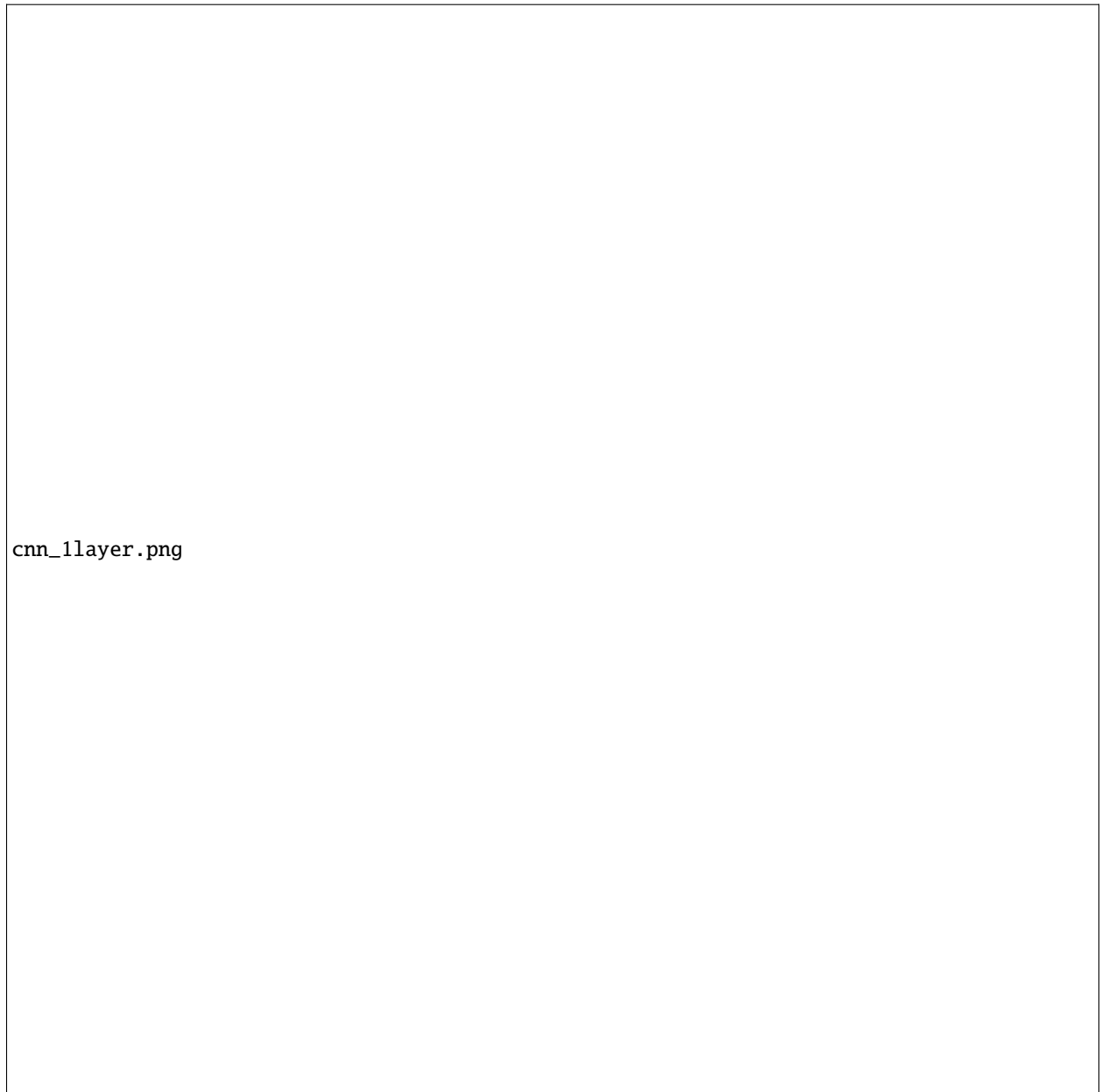


Fig. 3 Training curves, ROC curve, Precision-Recall curve, confusion matrix, and F1 vs. threshold for the 1-layer CNN.

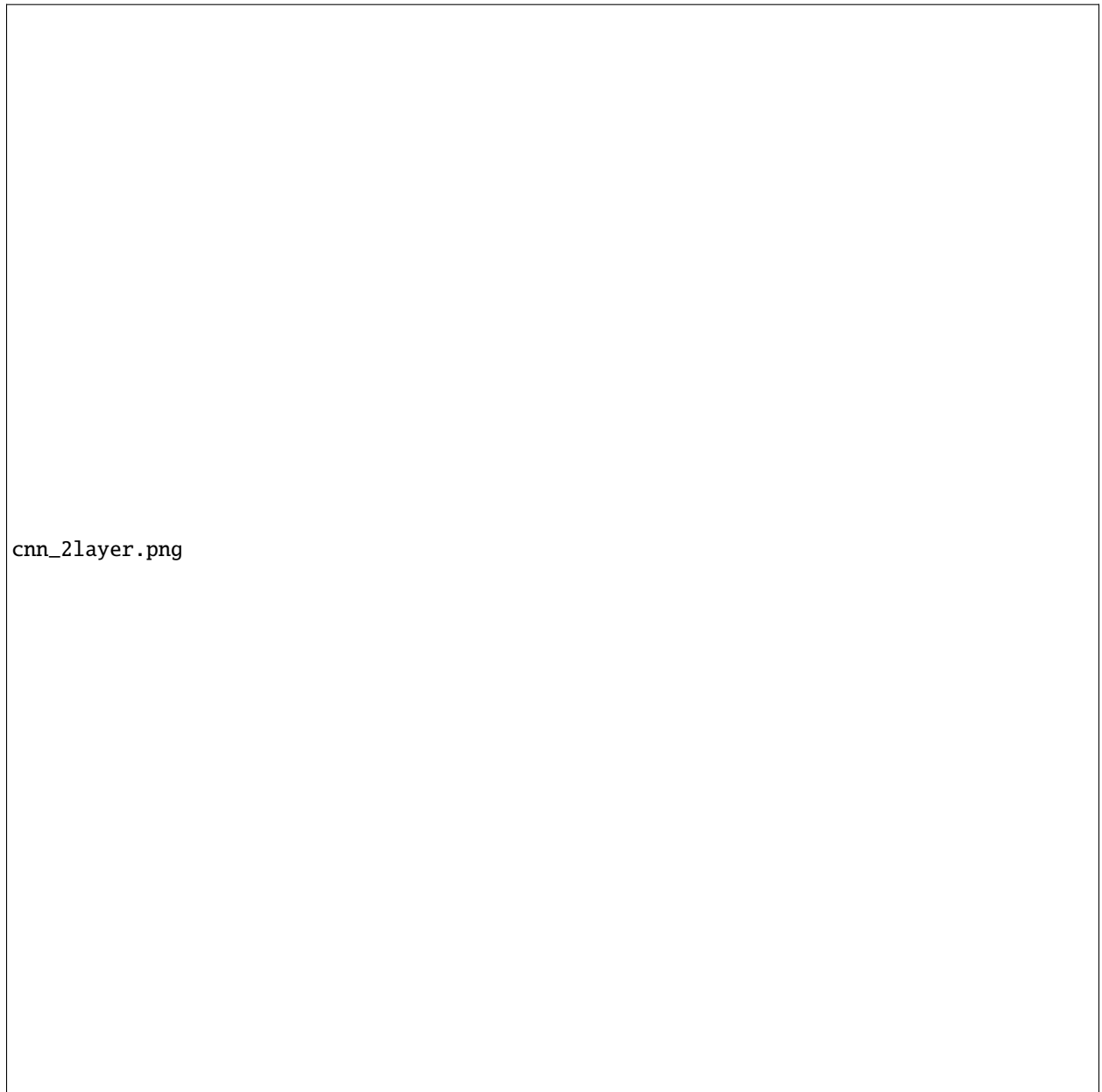


Fig. 4 Training curves, ROC curve, Precision-Recall curve, confusion matrix, and F1 vs. threshold for the 2-layer CNN.

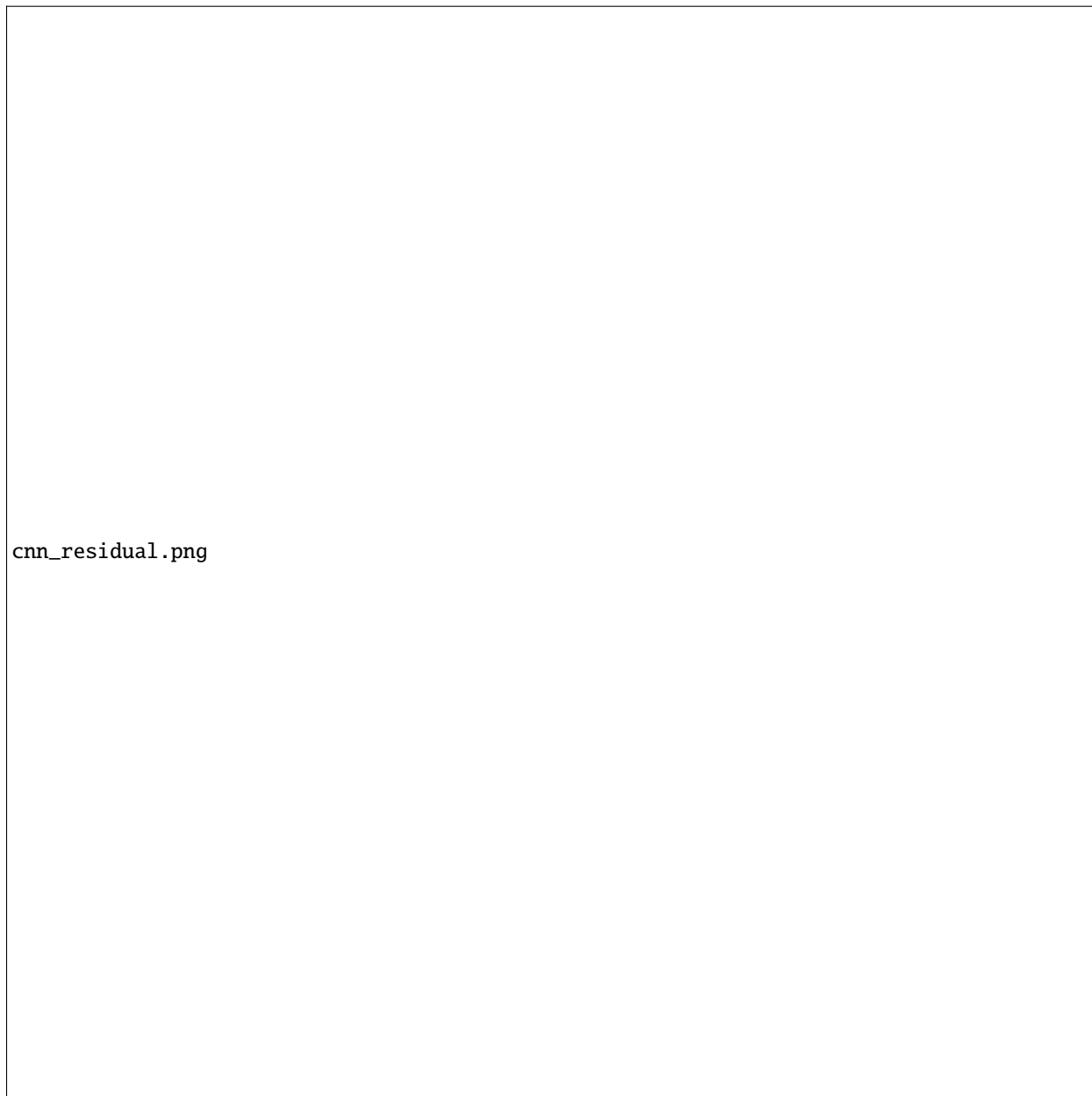


Fig. 5 Training curves, ROC curve, Precision-Recall curve, confusion matrix, and F1 vs. threshold for the residual CNN.